Discriminative Vision Transformer for Heterogeneous Cross-Domain Hyperspectral Image Classification

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***Abstract*— The transformer has been introduced in the hyperspectral image (HSI) classification, demonstrating outstanding capability in capturing global features compared to the convolutional neural network (CNN). However, the small-sample-size problem poses a significant challenge in practical HSI classification, especially in training the transformer. To tackle this issue, cross-domain transfer learning is adopted as a practical solution, which transfers the information from a source domain with abundant labeled samples to a target domain with limited labeled samples. This article proposes a novel transfer learning method for heterogeneous cross-domain HSI classification called cross-domain discriminative vision transformer (CD-DViT). This algorithm primarily contains three key contributions. First, source samples are mapped to the target domain through an encoder–decoder architecture, and the mapped source samples can be used to train the target classifier. Second, the cross-attention mechanism is utilized to construct two blocks for achieving the domainwise and classwise feature alignments (FAs), respectively. Specifically, the combination of the cross-attention mechanism with the domain discriminator aims to learn domain-invariant features, thereby facilitating domainwise alignment and alleviating domain shift. Third, knowledge distillation (KD) is adopted to learn more information from the target domain and assist in classifying target samples. Our experiments on three real-world cross-domain HSI datasets demonstrate the effectiveness of the proposed approach.**

***Index Terms*— Cross-domain classification, discriminative vision transformer (ViT), heterogeneous transfer learning, hyperspectral image (HSI), knowledge distillation (KD).**

NOMENCLATURE

$\mathcal{S}$ and $\mathcal{T}$	Source and target domains.
Q, **K**, and **V**	Query, key, and value matrices of self-attention.
$\mathbf{K}_{\mathcal{S}}^{L}$ and $\mathbf{V}_{\mathcal{S}}^{L}$	**K** and **V** from the last layer of DME.
$\mathbf{K}_{\mathcal{T}}^{i}$ and $\mathbf{V}_{\mathcal{T}}^{i}$	**K** and **V** from the i-layer of DMD.
$\mathbf{K}_{\mathcal{S}}^{m,i}$ and $\mathbf{V}_{\mathcal{S}}^{m,i}$	Mapped **K** and **V** from the i-layer of DMD.
$\mathrm{S}_{\mathcal{S}}$ and $\mathrm{S}_{\mathcal{T}}$	Source and target samples.
$\mathrm{E}_{\mathcal{S}}$ and $\mathrm{E}_{\mathcal{T}}$	Embedding source and target tokens.
$\mathrm{F}_{\mathcal{S}}$ and $\mathrm{F}_{\mathcal{T}}$	Source and target features.
$\mathrm{F}_{\mathcal{S}}^{m}$	Mapped source features.
$\mathrm{S}_{\mathcal{S}}^{m}$	Mapped source samples.
λ_1	Weight of cross-entropy loss.
λ_2	Weight of Kullback–Leibler divergence.
λ_3	Weight of distillation loss.
λ_4	Weight of domain discrimination loss.
τ	Temperature of distillation loss.
$\mathcal{L}^{\mathrm{cls}}$	Cross-entropy loss.
$\mathcal{L}^{\mathrm{KLD}}$	Kullback–Leibler divergence.
$\mathcal{L}^{\mathrm{KD}}$	Distillation loss.
$\mathcal{L}^{\mathrm{bin}}$	Domain discrimination loss.

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I. INTRODUCTION

PIXEL classification is a fundamental technique on hyperspectral images (HSIs). Over the past decades, numerous HSI classification methods have emerged, categorized into traditional machine learning- and deep learning-based methods. Traditional machine learning-based methods such as support vector machines (SVMs) [1], [2] and Markov random fields (MRFs) [3], [4] rely on manual feature extraction and lack adaptability in different datasets. In contrast, deep learning-based methods, convolutional neural networks (CNNs) [5], [6], transformer [7], [8], graph neural networks (GNNs) [9], [10], and recurrent neural networks (RNNs) [11], [12], can autonomously extract features.

Among all these deep learning-based methods, CNN is a dominant paradigm in computer vision (CV), excelling in extracting local features and achieving excellent performance in classification tasks [13]. However, the small convolution kernel size restricts the receptive field of CNN [14]. In contrast to CNN, the transformer can capture global features and establish long-range dependencies among sequential data. Initially introduced in natural language processing (NLP), the transformer has gradually become a widely used architecture in deep learning. Building upon the transformer, Dosovitskiy et al. [15] developed the vision transformer

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(ViT), applying the transformer to image classification. Consequently, numerous transformer-based techniques, particularly those based on ViT, have emerged in HSI classification. For instance, Mei et al. [16] proposed a group-aware hierarchical transformer (GAHT) to capture discriminative spectral–spatial features. Tu et al. [17] proposed a local semantic feature aggregation-based transformer (LSFAT) to learn multiscale features and capture locally spatial semantics adaptively. Ibanez et al. [18] proposed a masked autoencoding spectral–spatial transformer (MAEST) to explore the robust feature encoding and reconstruct the data for assisting in classification.

However, the above classification methods solely rely on a single dataset, termed single-domain HSI classification, and demand sufficient labeled training samples to achieve satisfactory performance. Unfortunately, manually labeling sufficient samples for practical applications is time-consuming, resulting in the small-sample-size problem in HSI classification. Besides, ViT has fewer image-specific inductive biases than CNN, necessitating more labeled training samples to train the model effectively [15]. These limitations pose a significant challenge in training ViT-based HSI classification. Consequently, cross-domain HSI classification has been introduced to transfer knowledge from a source domain with sufficient labeled samples to a target domain with limited labeled samples.

In cross-domain HSI classification, many proposed methods assume that the source and target domains are captured by the same sensor, sharing identical feature dimensions and labels [19]. This case is often referred to as homogeneous transfer learning, and various methods have been developed. For instance, Li et al. [20] utilized autoencoders to project source and target domains into a shared subspace, employing the multilevel maximum mean discrepancy (MMD) loss to alleviate domain shift. Ning et al. [21] introduced adaptive local discriminant analysis to extract discriminative features, which adopted distribution matching and subspace alignment for domain adaptation. Then, they designed an instance-to-instance contrastive learning framework based on category matching (CLCM) [22], introducing contrastive learning to extract domain-invariant and class-discriminative features.

Although the methods above have demonstrated outstanding performance, the source and target domains will have different feature dimensions and data distributions in most practical applications [19]. It poses challenges for homogeneous transfer learning. Thus, heterogeneous transfer learning has been introduced to alleviate the domain shift caused by different sensors. For instance, Li et al. [23] proposed an iterative reweighting heterogeneous transfer learning (IRHTL) framework, which iteratively learned a common subspace by weighted SVM and reweighted the source samples by the iterative reweighting strategy to establish relationships between two domains. Jiang et al. [24] proposed 3-D separable ResNet (3-D-SRNet), which adopted pretraining and fine-tuning strategies to achieve domain adaption. Utilizing the attention mechanism, Ye et al. [25] proposed a domain-invariant attention network (DIAN), constructing a domain-invariant attention block to alleviate the domain shift by projecting the source and target domains into a low-dimensional subspace. To make use of unlabeled samples in the target domain, Chen et al. [26] designed a semisupervised dual-dictionary nonnegative matrix factorization (SS-DDNMF) algorithm, which alleviates the domain shift by training two dictionaries for mapping two domains into a shared low-dimensional subspace.

Based on transfer learning, some researchers attempt to extend transformers from single-domain classification to cross-domain classification. For instance, Ran et al. [27] proposed deep transformer-based few-shot learning (DT-FSL), constructing transformer-based spatial attention and spectral query modules to explore potential relationships among nonlocal spatial samples and fully leverage spectral information. It primarily focused on feature alignment (FA) and neglected to explore cross-domain semantic relationships among objects. Thus, Cao et al. [28] proposed a cross-domain spectral semantic transformer few-shot learning (SSTFSL) method, which utilized the multihead self-attention (MSA) to extract the consistent semantic relations between source and target domains and guided the classification network. Moreover, some researchers have explored the combination of transformer and CNN to enrich the feature representations. Zhao et al. [29] employed CNN and transformer to achieve local and global FAs by a two-stage scheme. In addition, Peng et al. [30] designed a distribution aligner to reduce inner class distances and a domain discriminator to make convolutional transformer-based feature extractor capture domain-invariant features. These methods have typically employed few-shot learning frameworks to tackle the small-sample-size problem. However, a natural way is to utilize the encoder–decoder architecture of the transformer to address this issue. It provides a mapping-based approach to establish the relations between the source and target domains, enabling the mapped source samples to participate in the training of the target classifier.

This article introduces a novel approach, cross-domain discriminative ViT (CD-DViT), for heterogeneous transformer-based cross-domain HSI classification. To address the small-size-sample problem in ViT-based HSI classification, the encoder–decoder architecture is employed to map the source samples to the target domain, extending the available target training samples. Then, to narrow the inter-domain distribution dependencies between source and target domains, the discriminative cross-attention block (DCAB) is designed to encourage the feature extractor to capture domain-invariant representations. Furthermore, a classwise cross-attention block (CCAB) is designed for class-level FA, assigning low weights to features with low consistency and high weights to features with high consistency in a sample pair. Thus, the distribution discrepancies between two similar regions will be minimized, while those between two dissimilar regions will be neglected. In CD-DViT, three branches are designed for the transfer learning task: 1) the domain mapping (DM) branch maps the source samples to the target domain; 2) the FA branch aligns the source and target features using the cross-attention mechanism;

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and 3) the target classification (TC) branch classifies the target samples, including the extended ones. Besides, knowledge distillation (KD) is utilized to transfer more information from the TC branch to assist in classifying the target samples. The main contributions in this article are given as follows.

1) A novel triple-branch transformer-based approach is proposed for heterogeneous cross-domain HSI classification, which maps source samples to the target domain and effectively addresses the small-sample-size problem in ViT-based HSI classification.
2) Two blocks based on cross-attention mechanism are constructed to achieve domainwise and classwise FAs. Specifically, the domainwise FA block combines the cross-attention mechanism with the domain discriminator to learn domain-invariant features for alleviating domain shift.
3) KD is employed, with the TC branch serving as the teacher network and CCAB as the student network, to transfer more target knowledge to classify target samples.

The remaining sections of this article are organized as follows. Section II introduces the fundamentals of ViT, domain adaptation, and KD. Section III presents the architecture of CD-DViT and elaborates on its implementation details. The experimental results on three real-world cross-domain HSI datasets are shown in Section IV. Finally, the conclusion is drawn in Section V. The symbols used frequently in this article are listed in Nomenclature.

II. Related Works

A. Vision Transformer

The transformer [31] is an encoder–decoder architecture renowned for its impressive performance in tackling sequence-to-sequence tasks in NLP. Each layer of the encoder consists of two sublayers: MSA and feedforward network (FFN). Each layer of the decoder contains three sublayers: masked MSA (MMSA), multihead cross-attention (MCA), and FFN.

The transformer's success rests on its self-attention block, which excels at capturing global information and establishing long-range dependencies among patches. Fig. 1 illustrates the procedure of the self-attention block and the architecture of MSA in the transformer. In the self-attention block, input data will be projected into query matrix ($\mathbf{Q}$), key matrix ($\mathbf{K}$), and value matrix ($\mathbf{V}$) through three learnable linear projections: $\mathbf{W}_Q$, $\mathbf{W}_K$, and $\mathbf{W}_V$, respectively. The attention scores are computed via the dot product of $\mathbf{Q}$ and $\mathbf{K}$, and then, the softmax function is used. Afterward, the attention weights are derived by the dot product of attention scores and $\mathbf{V}$. It can be formulated as follows:

$$a = \text{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^T}{\sqrt{d}}\right)\mathbf{V} \quad (1)$$

where z signifies the attention weights and d signifies the dimensions of $\mathbf{K}$.

To capture diverse information from various representation subspaces, the MSA is employed to concatenate all attention weights from different heads, which can be formulated as follows:

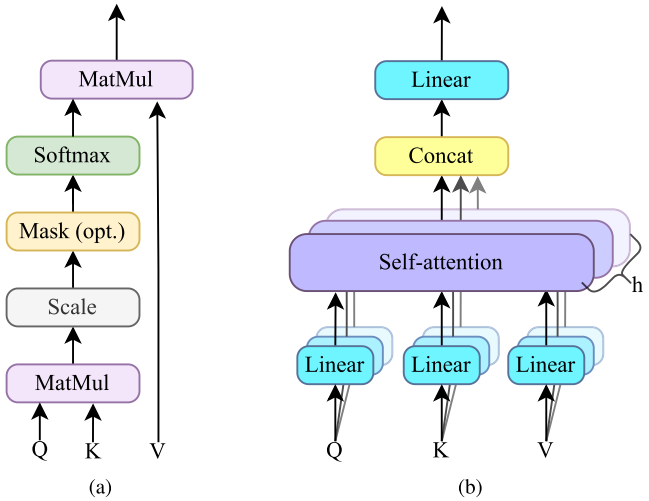


Fig. 1. Illustration of the self-attention block in ViT. (a) Procedure of self-attention block. (b) Architecture of MSA.

$$\text{MultiHead}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{Concat}(a_1, a_2, \ldots, a_h)\mathbf{W} \quad (2)$$

where h represents the number of heads, $\mathbf{W} \in \mathbb{R}^{hd_h \times d}$ denotes the weight matrix, and $d_h = (d/h)$ signifies the dimensions of each head. In particular, to prevent each patch in the decoder from attending to subsequent patches after obtaining attention scores, the MMSA is adopted to mask all illegal attention scores before using the softmax function.

Due to the remarkable success of the transformer, ViT was proposed, expanding its applicability to image classification tasks [15]. Unlike the standard transformer designed for 1-D sequential inputs, ViT demonstrates the ability to handle 2-D inputs. To process 2-D images, the input data $\mathbf{x} \in \mathbb{R}^{H \times W \times C}$ are initially reshaped into a sequence of nonoverlapping and flattened 2-D patches $\mathbf{x}_{\text{patch}} \in \mathbb{R}^{M \times (P^2 \times C)}$. Here, $H \times W$ represents the image resolution, C signifies the number of image channels, P^2 denotes the number of pixels of each patch, and $M = (HW/P^2)$ signifies the total number of patches. Subsequently, these patches are mapped to d dimensions using learnable linear embeddings, and positional embeddings are added to each patch to retain positional information.

Although the ViT has the powerful capability of feature extraction, it will perform poorly with limited training samples. To address this issue, the encoder–decoder architecture in the transformer offers a mapping-based approach to extend target training samples and provides the basis for implementing transfer learning. Based on these advantages, this article adopts ViT for feature extraction and utilizes the transformer to map source samples to the target domain.

B. Domain Adaptation

Domain adaptation primarily aims to leverage transferable knowledge from the source domain to aid in classifying target samples. It typically falls into two categories: discrepancy- and adversarial-based approaches [32].

The discrepancy-based method commonly narrows discrepancies between source and target domains by adding

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adaptation layers in a shared subspace of identical dimensions. Ye et al. [33] proposed a cross-domain variational autoencoder (CDVAE), which utilized a two-stream VAE layer to map two domains into a shared subspace and adopted graph regularization loss to alleviate domain shift. Zhang et al. [34] proposed the graph information aggregation cross-domain few-shot learning (Gia-CFSL) method, which employed two CNN-based mapping layers to map the features into identical dimensions and utilized graph optimal transport to achieve feature- and distribution-level domain alignments. Liu et al. [35] proposed a refined prototypical contrastive learning network for few-shot learning (RPCL-FSL), which combined the supervised contrastive learning with few-shot learning and applied a fusion training strategy to achieve FA.

Following the widespread adoption of generative adversarial networks (GANs) [36], adversarial-based methods have emerged as another prevalent approach to alleviate domain shift [37], [38]. Li et al. [39] proposed a deep cross-domain few-shot learning (DCFSL) method, which leverages a conditional adversarial domain adaptation strategy to achieve domain alignment. Meng et al. [40] designed a cycle auxiliary classifier generative adversarial network (Cycle-AC-GAN), establishing a two-way mapping between the source and target domains and adding land-cover classifiers within each discriminator to handle land-cover classification. Later, Ganin et al. [41] combined GAN with domain adaptation, proposing domain-adversarial neural network (DANN), which introduced gradient reversal layer (GRL) to optimize feature extractor and domain discriminator without alternating training by minimizing the domain discrimination loss. Tang et al. [42] constructed a domain discriminator to achieve domainwise FA and employed classifier determinacy disparity (CDD) to ensure the diversity of output probabilities and the discriminability of the generated features, where the domain discrimination loss and CDD loss should be maximized during training. The GRL allows both domainwise and classwise distributions to be aligned while minimizing these losses.

Based on the transformer, the cross-attention, derived from the self-attention, has demonstrated excellent capability in FA and has been employed in domain adaptation. For instance, Li et al. [43] designed a cross-attention map-based regularization (CAMR) method to constrain the generation of cross-attention maps for pulling the distributions of source and target samples with identical labels closer. Xu et al. [44] proposed a cross-domain transformer (CDTrans), which applied a cross-attention-based module to narrow distribution discrepancies by aligning features with high consistency between source and target samples. Motivated by CDTrans, Wang et al. [45] proposed the bidirectional cross-attention transformer (BCAT), creating a quadruple-branch architecture for combining self-attention and cross-attention with learning domain-invariant feature representations.

This article adopts the cross-attention to construct two blocks for achieving domainwise and classwise FAs.

C. Knowledge Distillation

KD is a fundamental knowledge transfer technique that has been widely applied in model compression and domain adaption. The main idea of KD is to leverage a teacher network with significant architecture to supervise the training of a student network with lightweight architecture, enabling the student network to learn more task-specific information from the teacher network [46]. Based on the different types of knowledge, KD can typically be divided into three categories: response-based, feature-based, and relation-based [47].

Response-based KD is a simple yet effective method that uses the final predictions of the teacher network to guide the training of the student network. For instance, Hu et al. [48] used KD to train a powerful feature extractor on the source domain and then fine-tuned it on the target domain. Yu et al. [49] constructed a dual KD module to achieve the decision-level knowledge integration, enabling the model to learn a powerful and effective classifier. Wu et al. [50] conducted KD on selecting the pseudolabels to improve their quality. Feature-based KD is an extension of the response-based KD, which uses the features from intermediate layers of the teacher network to guide the student network. For instance, Zhao et al. [51] combined metric learning with KD, matching the similarity of different hierarchical feature maps between two networks to transfer the knowledge. Gao et al. [52] proposed a triple-level distillation strategy, transferring channel-, feature- and logit-level information from the multiple layers in a well pretrained network into a student network. Mdrafi et al. [53] combined KD with activation-based attention maps to focus on the similarity between source and target distributions at different levels. Furthermore, relation-based KD explores the relationships between the samples or feature maps from different layers. For instance, Dong et al. [54] combined KD with clustering to reduce clustering error. Park et al. [55] proposed the distancewise and anglewise distillation losses, leveraging the mutual information in the teacher network to transfer the structure knowledge.

These methods applied the KD in domain adaptation, effectively transferring the knowledge between different domains. Thus, this article also adopts the KD to learn more transferable information.

III. Methodology

The architecture of CD-DViT is depicted in Fig. 2. It contains three branches: 1) DM branch maps heterogeneous data into identical dimensions and the source features into the target domain to extend the available target training samples; 2) FA branch aligns source and target domains by constructing two cross-attention-based blocks to achieve domainwise and classwise FAs; and 3) TC branch classifies the target samples, including the extended ones. The detailed descriptions are provided as follows.

A. DM Branch

A common challenge in HSI classification is the insufficiency of labeled training samples, which limits practical performance in deep learning-based models. To address this problem, the DM branch is designed to extract source and target features and then map the source features to the target domain, where the mapped samples can serve as the target

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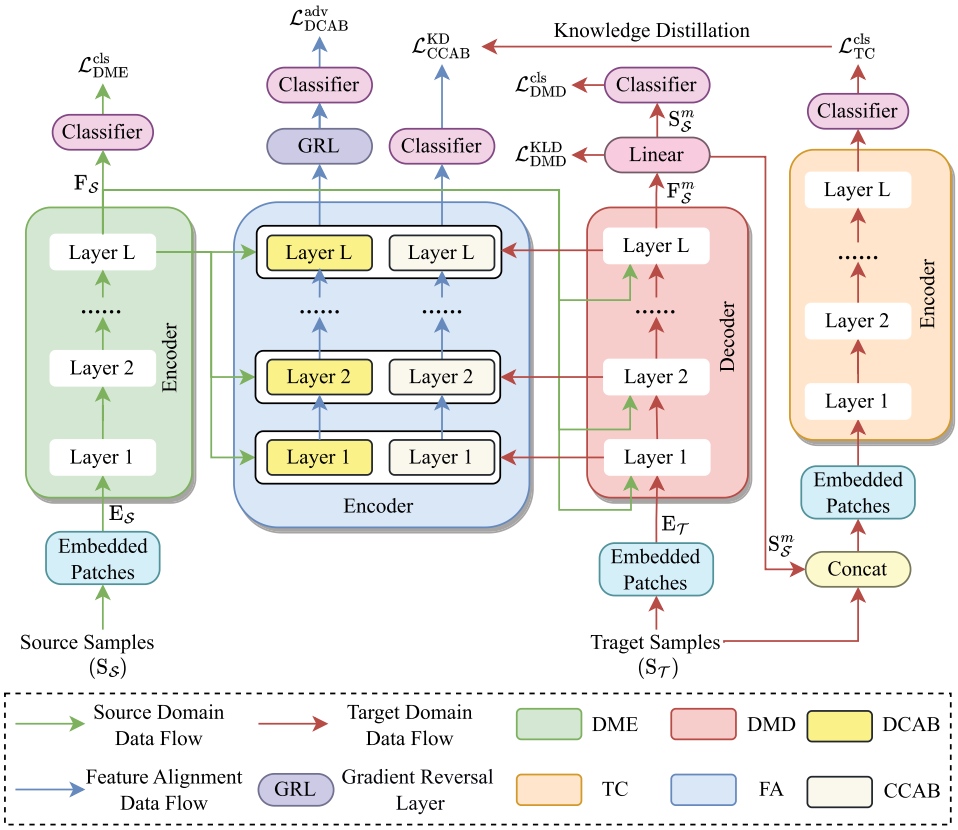


Fig. 2. Overall architecture of the proposed algorithm CD-DViT. The CD-DViT contains three branches: DM (which contains DME and DMD), FA, and TC. The source and target samples are first fed into DME and DMD to capture source and target features ($\mathrm{F}_{\mathcal{S}}$ and $\mathrm{F}_{\mathcal{T}}$), respectively. Meanwhile, the $\mathrm{F}_{\mathcal{S}}$ are fed into DMD to obtain mapped source features $\mathrm{F}_{\mathcal{S}}^{m}$. Then, all extracted features ($\mathrm{F}_{\mathcal{S}}$, $\mathrm{F}_{\mathcal{T}}$, and $\mathrm{F}_{\mathcal{S}}^{m}$) are fed into FA to achieve domainwise and classwise FAs. In DCAB, the GRL is employed to align the domain distributions while minimizing domain classification loss. Finally, $\mathrm{F}_{\mathcal{S}}^{m}$ is mapped to source feature space, obtaining $\mathrm{S}_{\mathcal{S}}^{m}$, and then, $\mathrm{S}_{\mathcal{T}}$ and $\mathrm{S}_{\mathcal{S}}^{m}$ are fed into TC to train the classification network.

training samples. Thus, the target domain will have sufficient available labeled samples for training.

The DM branch mainly consists of two modules: an L-layer DM encoder (DME) and an L-layer DM decoder (DMD). As depicted in Fig. 3(a), the DME captures source features, while the DMD captures target features and maps source features to the target domain.

For source and target samples ($\mathrm{S}_{\mathcal{S}}$ and $\mathrm{S}_{\mathcal{T}}$, where the subscript $\mathcal{S}$ and $\mathcal{T}$ indicate the source and target domains, respectively), we extract neighboring 3-D patches $\mathbf{x} \in \mathbb{R}^{P\times P\times C}$ centered at each pixel and flatten them into 2-D patches $\mathbf{x} \in \mathbb{R}^{P^2\times C}$. Then, as shown in Fig. 3(b), they are embedded into identical dimensions using two linear layers. In addition, a learnable class token (**CLS**) is embedded into the sequential tokens at the first position to learn classifiable information, and standard 1-D position embeddings (**P**) are added to retain positional information. Finally, the embedding source and target tokens ($\mathrm{E}_{\mathcal{S}}$ and $\mathrm{E}_{\mathcal{T}}$) can be obtained.

After getting $\mathrm{E}_{\mathcal{S}}$ and $\mathrm{E}_{\mathcal{T}}$, the $\mathrm{E}_{\mathcal{S}}$ are first fed into DME to capture source features $\mathrm{F}_{\mathcal{S}}$, and then, $\mathrm{F}_{\mathcal{S}}$ and $\mathrm{E}_{\mathcal{T}}$ will be fed into DMD. In DMD, the target features $\mathrm{F}_{\mathcal{T}}$ are captured by the MMSA sublayer, and then, the MCA sublayer receives $\mathrm{F}_{\mathcal{S}}$ and $\mathrm{F}_{\mathcal{T}}$ for mapping the $\mathrm{F}_{\mathcal{S}}$ to the target domain, obtaining mapped source features $\mathrm{F}_{\mathcal{S}}^{m}$ (where the superscript m indicates the mapped features). In the MCA sublayer, $\mathrm{F}_{\mathcal{S}}$ are projected as $\mathbf{K}_{\mathcal{S}}$ and $\mathbf{V}_{\mathcal{S}}$, while $\mathrm{F}_{\mathcal{T}}$ are projected as $\mathbf{Q}_{\mathcal{T}}$. To ensure class consistency, classification layers are added after DM, and cross-entropy losses based on land-cover classes are applied for training, denoting as $\mathcal{L}_{\mathrm{DME}}^{\mathrm{cls}}$ and $\mathcal{L}_{\mathrm{DMD}}^{\mathrm{cls}}$, respectively,

$$\mathcal{L}_{\mathrm{DME}}^{\mathrm{cls}} = -\frac{1}{N_{\mathcal{S}}}\sum_{i=1}^{N_{\mathcal{S}}}\sum_{j=1}^{C}\left(y_{ij}^{\mathcal{S}}\log\left(\hat{y}_{ij}^{\mathcal{S}}\right)\right) \tag{3}$$

$$\mathcal{L}_{\mathrm{DMD}}^{\mathrm{cls}} = -\frac{1}{N_{\mathcal{T}}}\sum_{i=1}^{N_{\mathcal{T}}}\sum_{j=1}^{C}\left(y_{ij}^{\mathcal{T}}\log\left(\hat{y}_{ij}^{\mathcal{T}}\right)\right) \tag{4}$$

where $N_{\mathcal{S}}$ and $N_{\mathcal{T}}$ represent the number of training samples in source and target domains, C is the number of land-cover classes, and y_{ij} and $\hat{y}_{ij}$ denote the ground-truth one-hot labels and predicted probability distributions of training samples.

It is worth noting that a linear layer is added before the classification layer in DMD to map the $\mathrm{F}_{\mathcal{S}}^{m}$ to the mapped source samples $\mathrm{S}_{\mathcal{S}}^{m}$ for restoring the dimensions before merging with $\mathrm{S}_{\mathcal{T}}$ to train the TC branch. To narrow distribution discrepancies between $\mathrm{S}_{\mathcal{T}}$ and $\mathrm{S}_{\mathcal{S}}^{m}$, the Kullback–Leibler Divergence (KLD) is employed

$$\mathcal{L}_{\mathrm{DMD}}^{\mathrm{KLD}} = \mathrm{KLD}(v,u) = \sum_{i=1}^{N} u_i \log\left(\frac{u_i}{v_i}\right) \tag{5}$$

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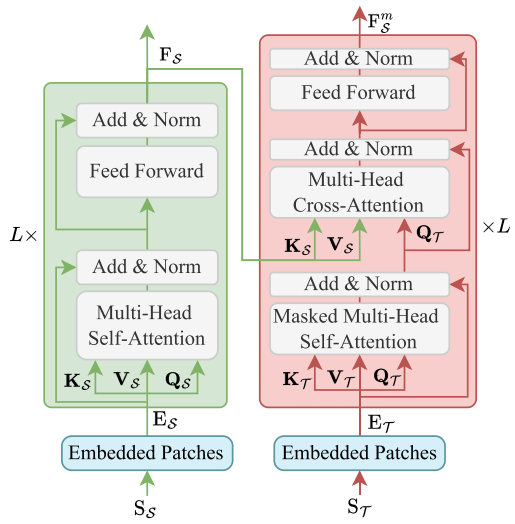


Fig. 3. Detailed architecture of the DM branch. (a) Architecture of the DM branch. (b) Detailed procedure of embedded patches.

where u_i and v_i represent the actual and predicted distributions of the ith sample.

B. FA Branch

Despite the availability of sufficient training target samples, distribution discrepancies still exist between the source and target domains. To address this issue, motivated by [44] and [56], the DCAB and the CCAB are designed in the FA branch to achieve domainwise and classwise FAs, respectively. The two blocks both rely on the cross-attention mechanism. Unlike the self-attention, whose $\mathbf{Q}$, $\mathbf{K}$, and $\mathbf{V}$ are from the same domain, the cross-attention takes them from different domains

$$\text{CrossAttn}(\mathbf{Q}_{\mathcal{S}}, \mathbf{K}_{\mathcal{T}}, \mathbf{V}_{\mathcal{T}}) = \text{softmax}\left(\frac{\mathbf{Q}_{\mathcal{S}}\mathbf{K}_{\mathcal{T}}^{T}}{\sqrt{d}}\right)\mathbf{V}_{\mathcal{T}}. \quad (6)$$

Thus, the features from different domains can be aligned. Besides, each block consists of an L-layer encoder.

The DCAB could encourage the DM branch to capture domain-invariant features to address domain distribution discrepancies. Inspired by [57], a set of randomly initialized $\mathbf{Q}_{\text{adv}}$ is employed to learn domain information, while $\mathbf{K}$ and $\mathbf{V}$ are obtained from the DME or DMD. They share a similar architecture for adversarial learning, with the DMD being illustrated as an example in Fig. 4. $\mathbf{Q}_{\text{adv}}$ and the cross-attention matrix can be computed as follows:

$$\mathbf{Q}_{\text{adv}}^{i} = \mathbf{Q}_{\text{adv}}^{i}\mathbf{W}_{F_d,Q}^{i} \quad (7)$$

$$\mathbf{Q}_{\text{adv}}^{i+1} = \text{CrossAttn}\left(\mathbf{Q}_{\text{adv}}^{i}, \mathbf{K}^{i}, \mathbf{V}^{i}\right) \quad (8)$$

where F_d denotes the DCAB.

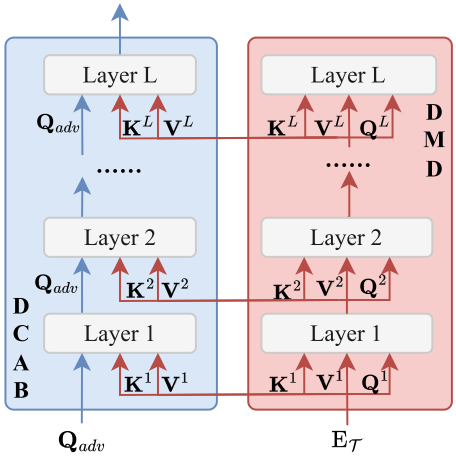


Fig. 4. Detailed architecture of DCAB.

For DME, $\mathbf{K}_{\mathcal{S}}^{L}$ and $\mathbf{V}_{\mathcal{S}}^{L}$ of the last layer are fed into each layer of DCAB, as only $F_{\mathcal{S}}^{L}$ of the last layer is fed into DMD for DM. Conversely, for the ith layer of DMD, $\mathbf{K}_{\mathcal{T}}^{i}$ and $\mathbf{V}_{\mathcal{T}}^{i}$ from MMSA sublayer are fed into the ith layer of DCAB. Similarly, $\mathbf{K}_{\mathcal{S}}^{m,i}$ and $\mathbf{V}_{\mathcal{S}}^{m,i}$ from the MCA sublayer are fed into the ith layer of DCAB. To train the DCAB, binary cross-entropy loss is employed

$$\begin{aligned}\mathcal{L}_{\text{DCAB}}^{\text{bin}} = -\frac{1}{N}\sum_{i=1}^{N}\Big(&\log D\left(\text{GRL}\left(F_d\left(\mathbf{K}_{\mathcal{T}}^{i}, \mathbf{V}_{\mathcal{T}}^{i}\right)\right)\right) \\ &+\log\left(1 - D\left(\text{GRL}\left(F_d\left(\mathbf{K}_{\mathcal{S}}^{L}, \mathbf{V}_{\mathcal{S}}^{L}\right)\right)\right)\right) \\ &+\log\left(1 - D\left(\text{GRL}\left(F_d\left(\mathbf{K}_{\mathcal{S}}^{m,i}, \mathbf{V}_{\mathcal{S}}^{m,i}\right)\right)\right)\right)\Big) \end{aligned} \quad (9)$$

where D signifies the domain classification layer in DCAB and GRL signifies gradient reversal layer. During training, the source samples $\text{S}_{\mathcal{S}}$ and mapped source samples $\text{S}_{\mathcal{S}}^{m}$ are labeled 0, and the target samples $\text{S}_{\mathcal{T}}$ are labeled 1.

It is important to emphasize that in general, minimizing the domain classification loss can make it easier to determine which domain a sample belongs to. However, by introducing a GRL and updating model parameters using reversed gradients, it becomes more challenging to differentiate the domain to which a sample belongs. In other words, it achieves the alignment of sample distributions between the two domains. In detail, GRL is a layer without trainable parameters. During the forward propagation, it is only used for identity transformation. During the backward propagation, the sign of gradient changes (multiplies it by a negative number), which takes from the succeeding layer and then feeds into the preceding layer. By introducing GRL, minimizing the label classification loss and maximizing the domain classification loss can be simultaneously achieved without alternating training.

In the FA branch, the CCAB is also designed to enhance the weights of features with high cross-domain consistency and weaken the weights of features with low cross-domain consistency. Fig. 5 depicts the detailed architecture of CCAB. In the ith layer of CCAB, $\text{F}_{\mathcal{S}}^{m,i}$, obtained from the output of the ith layer in DMD, are received and projected to $\mathbf{K}_{\mathcal{T}}^{i}$ and $\mathbf{V}_{\mathcal{T}}^{i}$, while $\text{F}_{\mathcal{S}}^{L}$, obtained from the output of the last layer of DME, are received and projected to $\mathbf{Q}_{\mathcal{S}}^{i}$. The computation of

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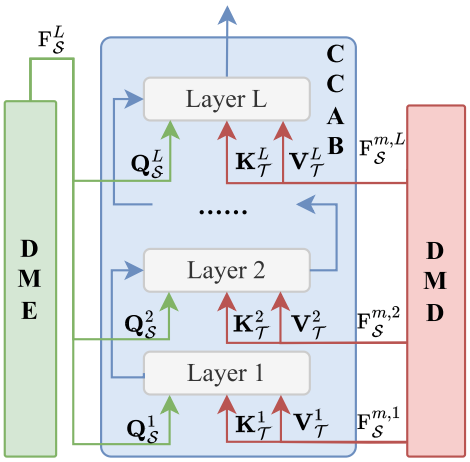


Fig. 5. Detailed architecture of CCAB.

the set of matrices $\{\mathbf{Q}_{\mathcal{S}}^{i}, \mathbf{K}_{\mathcal{T}}^{i}, \mathbf{V}_{\mathcal{T}}^{i}\}$ and the cross-attention matrix A_{CCAB}^{i} are given as follows:

$$\mathbf{Q}_{\mathcal{S}}^{i} = \mathrm{F}_{\mathcal{S}}^{L}\mathbf{W}_{\mathcal{T},Q}^{i} \tag{10}$$

$$\mathbf{K}_{\mathcal{T}}^{i} = \mathrm{F}_{\mathcal{S}}^{m,i}\mathbf{W}_{\mathcal{T},K}^{i} \tag{11}$$

$$\mathbf{V}_{\mathcal{T}}^{i} = \mathrm{F}_{\mathcal{S}}^{m,i}\mathbf{W}_{\mathcal{T},V}^{i} \tag{12}$$

$$A_{\mathrm{CCAB}}^{i} = \mathrm{CrossAttn}\left(\mathbf{Q}_{\mathcal{S}}^{i}, \mathbf{K}_{\mathcal{T}}^{i}, \mathbf{V}_{\mathcal{T}}^{i}\right). \tag{13}$$

Besides, to enhance the class discriminability of $\mathrm{S}_{\mathcal{S}}^{m}$, KD is used to transfer more knowledge from the TC branch (the teacher network) to guide the training of the CCAB (the student network). Unlike the other methods, the KD leverages the teacher network to provide more detailed target knowledge of land-cover classes for the student network, which can enhance classification performance in the target domain. The loss can be formulated as follows:

$$\mathcal{L}_{\mathrm{CCAB}}^{\mathrm{KD}} = \mathrm{KLD}(p(z_s, \tau), p(z_t, \tau)) \tag{14}$$

where z_s and z_t represent the output probabilities of the student and teacher networks. Besides, p is the soft labels of KD, which can be defined as $p(z^i, \tau) = (\exp(z^i/\tau)/\sum_j \exp(z^j/\tau))$, z^i is the logit of the ith class, and τ is the temperature to control the importance of each soft label.

C. TC Branch and Total Loss

The TC branch is designed to train the classifier on the target domain, consisting of an L-layer encoder and receiving the $\mathrm{S}_{\mathcal{T}}$ and $\mathrm{S}_{\mathcal{S}}^{m}$ for training. Cross-entropy loss $\mathcal{L}_{\mathrm{TC}}^{\mathrm{cls}}$ based on land-cover classes is applied for training

$$\mathcal{L}_{\mathrm{TC}}^{\mathrm{cls}} = -\frac{1}{N_{\mathcal{S}} + N_{\mathcal{T}}} \sum_{i=1}^{N_{\mathcal{S}}+N_{\mathcal{T}}} \sum_{j=1}^{C} \left(y_{ij} \log\left(\hat{y}_{ij}\right)\right). \tag{15}$$

Besides, the encoders among all branches share the classification layer to optimize training parameters and ensure class consistency. In summary, the total loss of the proposed algorithm is given as follows:

$$\mathcal{L}_{\mathrm{total}} = \mathcal{L}_{\mathrm{TC}}^{\mathrm{cls}} + \lambda_1\left(\mathcal{L}_{\mathrm{DME}}^{\mathrm{cls}} + \mathcal{L}_{\mathrm{DMD}}^{\mathrm{cls}}\right) + \lambda_2\mathcal{L}_{\mathrm{DMD}}^{\mathrm{KLD}}$$

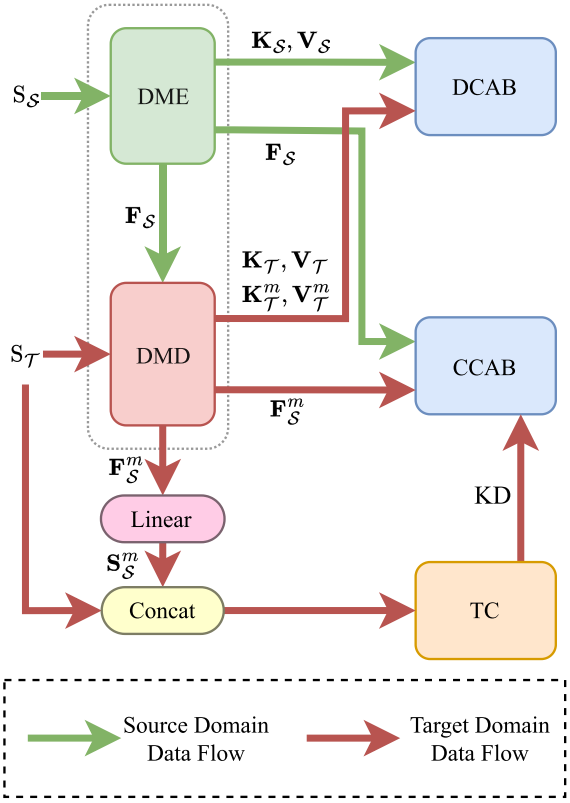


Fig. 6. Flowchart of data in CD-DViT.

$$+ \lambda_3\mathcal{L}_{\mathrm{CCAB}}^{\mathrm{KD}} + \lambda_4\mathcal{L}_{\mathrm{DCAB}}^{\mathrm{bin}} \tag{16}$$

where λ_1–λ_4 represent hyperparameters used to balance the loss functions.

D. Training Details

As depicted in Fig. 6, the detailed training procedures of CD-DViT are given as follows.

1) The DM receives $\mathrm{S}_{\mathcal{S}}$ and $\mathrm{S}_{\mathcal{T}}$, calculating the $\mathcal{L}_{\mathrm{DME}}^{\mathrm{cls}}$, $\mathcal{L}_{\mathrm{DMD}}^{\mathrm{cls}}$, and $\mathcal{L}_{\mathrm{DMD}}^{\mathrm{KLD}}$.
2) The DCAB and CCAB receive the output of DME and DMD, prompting the DM branch to learn domain-invariant features and making the distributions with identical land-cover classes closer, but only $\mathcal{L}_{\mathrm{DCAB}}^{\mathrm{bin}}$ is calculated.
3) $\mathrm{S}_{\mathcal{T}}$ and $\mathrm{S}_{\mathcal{S}}^{m}$ are fed into the TC branch; $\mathcal{L}_{\mathrm{TC}}^{\mathrm{cls}}$ and $\mathcal{L}_{\mathrm{CCAB}}^{\mathrm{KD}}$ are calculated. KD is applied to enhance the class discriminability of $\mathrm{S}_{\mathcal{S}}^{m}$.

In brief, the DM branch is used to extract coarse-grained source and target features. Then, to alleviate the domain shift, the FA branch is applied to perform fine-grained cross-domain FA. Furthermore, based on cross-domain feature consistency, the KD is used between the TC branch and CCAB to improve the class discriminability of mapped source features. Algorithm 1 summarizes more detailed information, including the data flow through the model and the integration of branches.

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Algorithm 1 CD-DViT Algorithm

Input: Source and target training samples with their corresponding labels, the number of training epoch N_{epoch}, hyperparameters λ_1, λ_2, λ_3, λ_4 and τ.

Output: Predicted labels of test samples.

Match the source and target samples with identical land-cover classes and construct $\mathrm{S}_{\mathcal{S}}$ and $\mathrm{S}_{\mathcal{T}}$.
Randomly initialize $\mathbf{Q}_{\text{adv}}$ of DCAB.
for epoch $\leftarrow 0$ to N_{epoch} **do**
  Extract coarse-grained source and target features:
    1) Capture $\mathrm{F}_{\mathcal{S}}^{L}$, $\mathbf{Q}_{\mathcal{S}}^{L}$, $\mathbf{K}_{\mathcal{S}}^{L}$, $\mathbf{V}_{\mathcal{S}}^{L}$ through DME($\mathrm{S}_{\mathcal{S}}$), computing $\mathcal{L}_{\text{DME}}^{\text{cls}}$ by Eq. (3).
    2) Capture $\{\mathrm{F}_{\mathcal{S}}^{m,i}, \mathbf{Q}_{\mathcal{T}}^{i}, \mathbf{K}_{\mathcal{T}}^{i}, \mathbf{V}_{\mathcal{T}}^{i}\}_{i=1}^{L}$ through DMD($\mathrm{S}_{\mathcal{T}}$, $\mathrm{F}_{\mathcal{S}}^{L}$), computing $\mathcal{L}_{\text{DMD}}^{\text{cls}}$ and $\mathcal{L}_{\text{DMD}}^{\text{KLD}}$ by Eq. (4) and (5).
  Perform fine-grained cross-domain FA:
    1) Compute $\mathcal{L}_{\text{DCAB}}^{\text{bin}}$ for DCAB($\mathbf{Q}_{\text{adv}}$; $\mathbf{K}_{\mathcal{S}}^{L}$, $\mathbf{V}_{\mathcal{S}}^{L}$; $\mathbf{K}_{\mathcal{T}}^{i}$, $\mathbf{V}_{\mathcal{T}}^{i}$; $\mathbf{K}_{\mathcal{S}}^{m,i}$, $\mathbf{V}_{\mathcal{S}}^{m,i}$) by Eq. (9).
    2) Map $\mathrm{F}_{\mathcal{S}}^{L}$ to $\mathbf{Q}_{\mathcal{S}}^{i}$, and $\mathrm{F}_{\mathcal{S}}^{m,i}$ to ($\mathbf{K}_{\mathcal{T}}^{i}$, $\mathbf{V}_{\mathcal{T}}^{i}$) by Eq. (10) to (12).
    3) Compute A_{CCAB}^{i} through CCAB($\mathbf{Q}_{\mathcal{S}}^{i}$, $\mathbf{K}_{\mathcal{T}}^{i}$, $\mathbf{V}_{\mathcal{T}}^{i}$) by Eq. (13).
  Enhance the class discriminability of $\mathrm{S}_{\mathcal{S}}$:
    1) Obtain predicted probability distributions from TC($\mathrm{S}_{\mathcal{T}}$, $\mathrm{S}_{\mathcal{S}}^{m}$), computing $\mathcal{L}_{\text{TC}}^{\text{cls}}$ by Eq. (15).
    2) Compute $\mathcal{L}_{\text{CCAB}}^{\text{KD}}$ for KD(A_{CCAB}, predicted probability distributions) by Eq. (14).
  Optimize the total loss defined by Eq. (16) to train the CD-DViT.
end for
Input the test samples for classification.

IV. Experiments

A. Experimental Dataset Descriptions

To validate the effectiveness of our proposed algorithm, we conduct experiments on three real-world cross-domain HSI datasets, including the EHangzhou-RPaviaHR dataset, the RPaviaC-DPaviaU dataset, and the Houston2018-Houston2013 dataset. The detailed information is described as follows.

The first cross-domain dataset is *EHangzhou-RPaviaHR Dataset*, which consists of the EO-1 Hangzhou (EHangzhou) scene and the ROSIS Pavia HR (RPaviaHR) scene. The EHangzhou scene serves as the source domain, captured by the EO-1 Hyperion hyperspectral sensor over Hangzhou City, Zhejiang, China. Besides, the RPaviaHR scene serves as the target domain, collected by the ROSIS HSI sensor over Pavia City, Italy. The source domain (EHangzhou) contains 198 spectral bands and 590 × 230 spatial pixels, and the target domain (RPaviaHR) contains 102 spectral bands and 1400 × 512 spatial pixels. Three land-cover classes are shared between the two domains. The name of land-cover classes and the number of samples per class are listed in Table I. In addition, the false-color and ground-truth maps are displayed in Fig. 7.

The second cross-domain dataset is *RPaviaC-DPaviaU Dataset*, which consists of the ROSIS Pavia Center (RPaviaC) scene and the DAIS Pavia University (DPaviaU) scene. The RPaviaC scene serves as the source domain, captured by the ROSIS HSI sensor over the center of Pavia City, Italy. Besides, the DPaviaU scene serves as the target domain, collected by the DAIS sensor over Pavia University, Italy. The source domain (RPaviaC) contains 102 spectral bands and 1096 × 715 spatial pixels, and the target domain (DPaviaU) contains 72 spectral bands and 243 × 243 spatial pixels. Five land-cover classes are shared between the two domains. The name of land-cover classes and the number of samples per class are listed in Table II. In addition, the false-color and ground-truth maps are shown in Fig. 8.

The third cross-domain dataset is *Houston2018-Houston2013 Dataset*, which consists of two scenes captured by the different sensors over the campus area and the neighboring urban area of the University in 2018 (Houston2018) and 2013 (Houston2013), respectively. The

TABLE I
NUMBER OF LABELED SAMPLES IN EACH LAND-COVER CLASS WITHIN THE EHANGZHOU-RPAVIAHR DATASET

#	Name	EHangzhou	RPaviaHR
C1	Water	18043	22525
C2	Ground/Building	77450	145416
C3	Vegetation	40207	22961
	Total	135700	190902

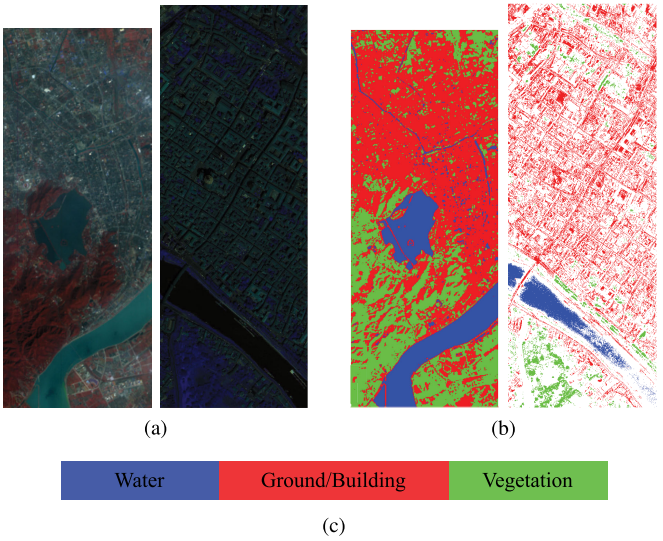


Fig. 7. False-color maps and ground-truth maps of the EHangzhou-RPaviaHR dataset. (a) False-color maps with bands 25, 50, and 100. (b) Ground-truth maps. (c) Three shared classes between two domains. The top maps are from the source domain (EHangzhou), while the bottom ones are from the target domain (RPaviaHR).

TABLE II
NUMBER OF LABELED SAMPLES IN EACH LAND-COVER CLASS WITHIN THE RPAVIAC-DPAVIAU DATASET

#	Name	RPaviaC	DPaviaU
C1	Trees	7598	135
C2	Meadows	3090	137
C3	Soil	6584	110
C4	Asphalt	9248	137
C5	Bitumen	7287	107
	Total	33807	626

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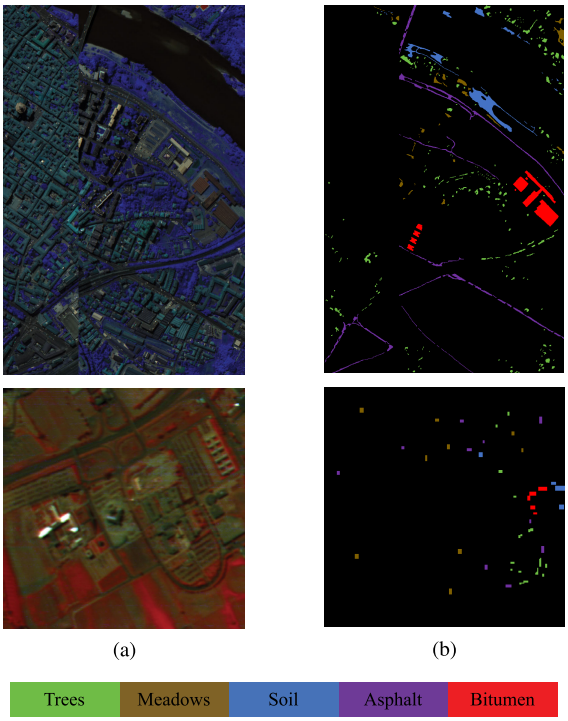


Fig. 8. False-color maps and ground-truth maps of the RPaviaC-DPaviaU dataset. (a) False-color maps with bands 25, 50, and 70. (b) Ground-truth maps. (c) Five shared classes between two domains. The top maps are from the source domain (RPaviaC), while the bottom ones are from the target domain (DPaviaU).

TABLE III
NUMBER OF LABELED SAMPLES IN EACH LAND-COVER CLASS WITHIN THE HOUSTON2018-HOUSTON2013 DATASET

#	Name	Houston2018	Houston2013
C1	Healthy grass	9799	1251
C2	Stressed grass	32502	1254
C3	Trees	13595	1244
C4	Water	266	325
C5	Residential buildings	39772	1268
C6	Commercial buildings	223752	1244
C7	Road	45866	1252
	Total	365552	7838

Houston2018 scene serves as the source domain, containing 48 spectral bands and 601 × 2384 spatial pixels, while the Houston2013 scene serves as the target domain, containing 144 spectral bands and 349 × 1905 spatial pixels. Seven shared land-cover classes are selected between the two domains. The name of land-cover classes and the number of samples per class are listed in Table III. In addition, the false-color and ground-truth maps are presented in Fig. 9.

B. Experimental Settings

1) *Implementation Details:* In cross-domain HSI classification, the source domain typically contains an abundance of labeled training samples, and the target domain contains a limited number of labeled training samples. For our experiments, we randomly select 50 labeled training samples per class from the source domain and five labeled training samples per class from the target domain. To conform to the input requirements of the transformer, we match the source and target samples with identical land-cover classes to obtain source–target sample pairs, resulting in 250 sample pairs per land-cover class. The remaining samples in the target domain are regarded as test samples. It is crucial to note that the source training samples are only utilized during the training procedure, while the testing is conducted solely on the target domain.

2) *Compared Algorithms:* We compare the classification performance of nine different HSI classification methods, including deep pyramidal residual networks (DPyResNet) [5], hybrid spectral CNN (HybridSN) [6], ViT [15], SpectralFormer [14], 3-D-SRNet [24], Cycle-AC-GAN [40], Gia-CFSL [34], DCFSL [39], and RPCL-FSL [35]. The DPyResNet, HybridSN, ViT, and SpectralFormer are single-domain algorithms, and 3-D-SRNet, Cycle-AC-GAN, Gia-CFSL, DCFSL, and RPCL-FSL are cross-domain algorithms. Moreover, the single-domain algorithms are trained only on the target domain, while the cross-domain algorithms are trained on the source and target domains. Table IV provides the settings of loss functions and main parameters for each method, and the detailed descriptions are given as follows.

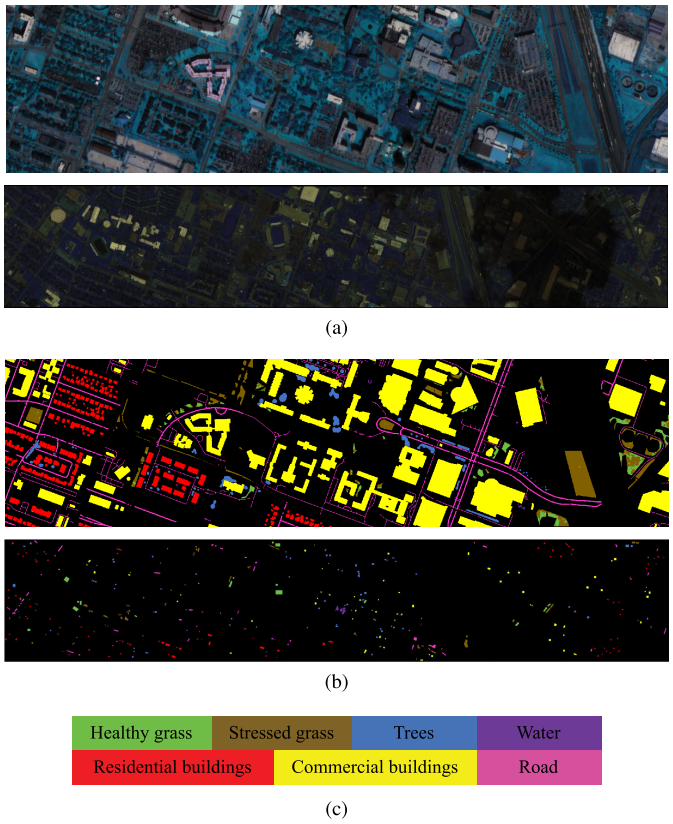


Fig. 9. False-color maps and ground-truth maps of the Houston2018-Houston2013 dataset. (a) False-color maps with bands 25, 50, and 100. (b) Ground-truth maps. (c) Seven shared classes between two domains. The top maps are from the source domain (Houston2018), while the bottom ones are from the target domain (Houston2013).

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TABLE IV
COMPARED ALGORITHMS AND MAIN PARAMETER SETTINGS

Algorithm	Settings
DPyResNet [5]	Patch size: 9×9 Learning rate: 0.001 Number of training epochs: 1000
HybridSN [6]	Patch size: 25×25 Learning rate: 0.001 Number of training epochs: 1000
ViT [15]	Patch size: 3×3 Learning rate: 0.0001 Number of training epochs: 1000 Embedding dimensions: 768 Number of layers: 6
SpectralFormer [14]	Patch size: 7×7 Optimizer: Adam Learning rate: 5×10^{-4} Number of training epochs: 1000
3-D-SRNet [24]	Patch size: 9×9 Learning rate: 0.01 Number of pre-training epochs: 1000 Number of fine-tuning epochs: 1000
Cycle-AC-GAN [40]	Patch size: 1×1 Learning rate: 0.01 Number of training epochs: 1000
Gia-CFSL [34]	Patch size: 9×9 Learning rate: 0.001 Number of training epochs: 1000
DCFSL [39]	Patch size: 9×9 Learning rate: 0.001 Number of training epochs: 1000
RPCL-FSL [35]	Patch size: 9×9 Learning rate: 0.001 Number of training epochs: 1000
CD-DViT	Patch size: 3×3 Learning rate: 0.0001 Number of training epochs: 1000 Embedding dimensions: 768 Number of layers: 6

a) *DPyResNet* is a CNN-based single-domain HSI classification algorithm proposed by Paoletti et al. [5]. It is based on the pyramidal bottleneck residual units and can capture more spectral–spatial features as the network becomes deeper.
b) *HybridSN* is a CNN-based single-domain HSI classification algorithm proposed by Roy et al. [6]. It employs PCA for spectral dimension reduction and combines 2-D and 3-D CNNs to extract spectral–spatial features, achieving satisfactory performance with sufficient training samples.
c) *ViT* is a transformer-based single-domain image classification algorithm proposed by Dosovitskiy et al. [15], achieving excellent performance in image classification tasks.
d) *SpectralFormer* is a transformer-based single-domain image classification algorithm proposed by Hong et al. [14], which designs the cross-layer skip connection and can learn spectral information from neighboring bands.
e) *3-D-SRNet* is a cross-domain HSI classification algorithm proposed by Jiang et al. [24]. It first pretrains on the source domain with sufficient labeled source samples and then fine-tunes the target domain for enhanced performance.
f) *Cycle-AC-GAN* is a cross-domain HSI classification algorithm proposed by Meng et al. [40]. It designs a two-way mapping to establish the relations between the source and target domains. The discriminators contain binary classifiers for domain classification and auxiliary classifiers for classifying land-cover classes.
g) *Gia-CFSL* is a cross-domain HSI classification method proposed by Zhang et al. [34], which leverages GNN to aggregate graph information and designs a graph optimal transport method for feature- and distribution-level domain alignments.
h) *DCFSL* is a cross-domain HSI classification method proposed by Li et al. [39], which leverages conditional adversarial domain adaptation strategy to learn a feature embedded space and reduce distribution discrepancies between two domains.
i) *RPCL-FSL* is a cross-domain HSI classification method proposed by Liu et al. [35], which combines the few-shot learning with supervised contrastive learning and applies triple constraints to learn high-quality support set prototypes.

3) *Evaluation Metrics:* Three standard metrics, including overall accuracy (OA), average accuracy (AA), and kappa coefficient (κ), are chosen to evaluate the performance of classification for each method. It is worth noting that all three metrics are derived from the average of ten independent runs with random sampling.

C. Experiments of Hyperparameter Settings

In CD-DViT, the total loss function contains multiple loss functions, with five hyperparameters requiring in-depth discussion. We set different values for $\lambda_1 \in \{0.1, 0.2, \ldots, 1.0\}$, $\lambda_2, \lambda_3 \in \{0.1, 0.3, \ldots, 1.1\}$, $\lambda_4 \in \{0.8, 0.9, \ldots, 1.2\}$, and $\tau \in \{1, 3, \ldots, 15\}$. To analyze the impact of these parameters on performance, we adopt OA to evaluate the performance of the proposed algorithm on three different HSI datasets. The experimental results are illustrated in Fig. 10, and the analyses of the hyperparameters are presented as follows.

1) Observing the results depicted in Fig. 10, it is evident that experimental results are insensitive to changes in hyperparameters. It indicates that our proposed model is robust to different hyperparameter settings.
2) The specific hyperparameter settings of each dataset are given as follows.
   a) For the EHangzhou-RPaviaHR dataset, the hyperparameters are set to $\lambda_1 = 0.3$, $\lambda_2 = 0.5$, $\lambda_3 = 0.7$, $\lambda_4 = 1.1$, and $\tau = 13.0$.
   b) For the RPaviaC-DPaviaU dataset, the hyperparameters are set to $\lambda_1 = 1.0$, $\lambda_2 = 0.7$, $\lambda_3 = 1.1$, $\lambda_4 = 1.0$, and $\tau = 3.0$.
   c) For the Houston2018-Houston2013 dataset, the hyperparameters are set to $\lambda_1 = 1.0$, $\lambda_2 = 0.9$, $\lambda_3 = 0.9$, $\lambda_4 = 1.0$, and $\tau = 3.0$.

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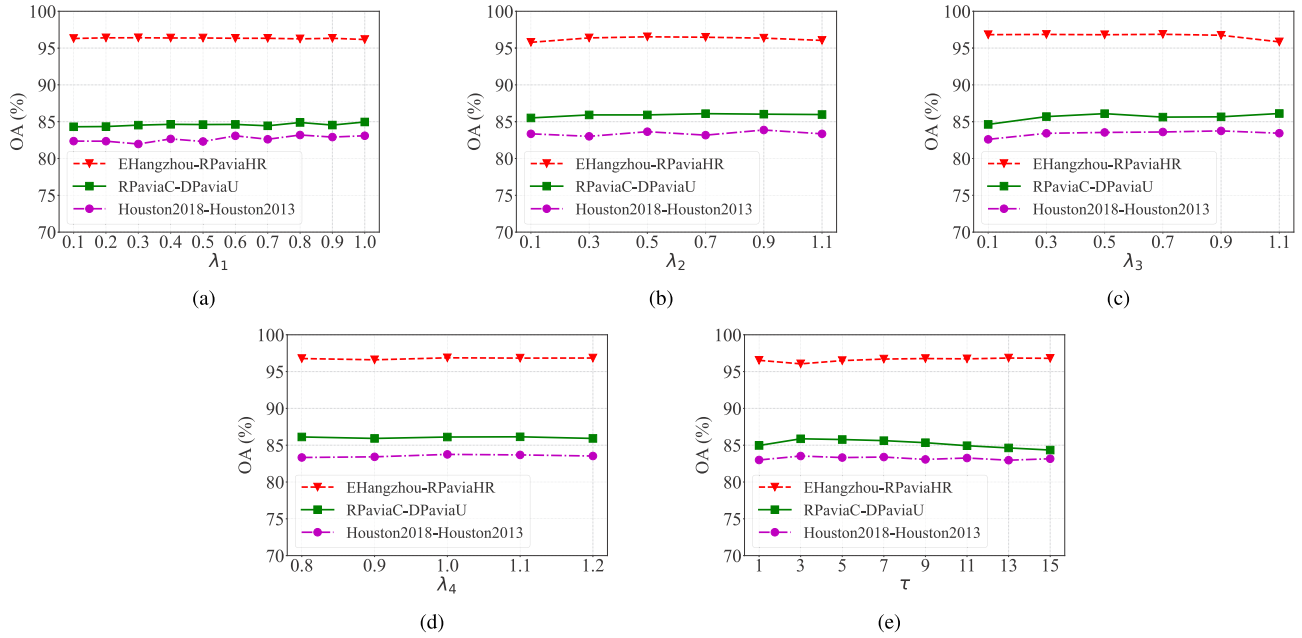


Fig. 10. Impact of varying hyperparameters on OA among three datasets. (a) λ_1. (b) λ_2. (c) λ_3. (d) λ_4. (e) τ.

TABLE V

OA, AA, AND κ OF EACH METHOD ON THE EHANGZHOU-RPAVIAHR DATASET

Method	OA(%)	AA(%)	$\kappa(\times 100)$
DPyResNet	80.88	77.13	64.69
HybridSN	84.38	87.65	69.62
ViT	91.87	96.37	82.94
SpectralFormer	94.13	97.02	87.48
3-D-SRNet	87.86	92.85	74.78
Cycle-AC-GAN	90.08	95.49	78.92
Gia-CFSL	94.47	97.29	87.72
DCFSL	93.02	96.24	84.40
RPCL-FSL	90.41	94.99	79.28
CD-DViT	**96.83**	**98.58**	**92.76**

TABLE VI

OA, AA, AND κ OF EACH METHOD ON THE RPAVIAC-DPAVIAU DATASET

Method	OA(%)	AA(%)	$\kappa(\times 100)$
DPyResNet	70.15	74.99	62.59
HybridSN	68.34	68.95	60.37
ViT	82.61	82.63	78.23
SpectralFormer	76.02	76.55	70.02
3-D-SRNet	74.04	74.05	67.44
Cycle-AC-GAN	75.06	74.83	68.77
Gia-CFSL	78.70	78.73	73.35
DCFSL	82.98	83.66	78.71
RPCL-FSL	80.58	81.11	75.70
CD-DViT	**86.11**	**86.25**	**82.61**

TABLE VII

OA, AA, AND κ OF EACH METHOD ON THE HOUSTON2018-HOUSTON2013 DATASET

Method	OA(%)	AA(%)	$\kappa(\times 100)$
DPyResNet	74.39	75.04	69.74
HybridSN	62.98	66.19	56.39
ViT	82.39	83.19	78.21
SpectralFormer	77.95	76.66	73.92
3-D-SRNet	75.93	76.24	71.62
Cycle-AC-GAN	78.97	79.31	75.19
Gia-CFSL	80.56	80.96	77.02
DCFSL	78.22	77.74	74.27
RPCL-FSL	74.02	73.74	69.29
CD-DViT	**83.75**	**84.48**	**80.80**

D. Comparison With Different Algorithms

The analysis of the experiments commences with the EHangzhou-RPaviaHR dataset. Table V displays the results of all algorithms. The analyses are given as follows.

1) Experimental results in Table V indicate poor performance of the DPyResNet and HybridSN methods due to limited training samples. Both ViT and SpectralFormer outperform DPyResNet, HybridSN, 3-D-SRNet, Cycle-AC-GAN, and RPCL-FSL, demonstrating the superiority in feature extraction of transformers. In addition, the transformer-based method CD-DViT exhibits better performance than 3-D-SRNet, Cycle-AC-GAN, Gia-CFSL, DCFSL, and RPCL-FSL. These results show the superior feature extraction capabilities of transformer-based models over CNN-based models.
2) As seen in Table V, CD-DViT outperforms ViT and SpectralFormer, indicating effective knowledge transfer from the source domain to the target domain.
3) Among all the compared algorithms, CD-DViT demonstrates the best classification performance.

After analyzing the EHangzhou-RPaviaHR dataset, the experimental analyses are conducted on the RPaviaC-DPaviaU dataset, with results detailed in Table VI. The analyses are given as follows.

1) Results in Table VI reveal that DPyResNet and HybridSN still perform poorly, while ViT and SpectralFormer perform better than 3-D-SRNet and Cycle-AC-GAN. Furthermore, CD-DViT still outperforms other transfer learning methods. These results demonstrate the better feature extraction

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TABLE VIII
ABLATION STUDIES OF EACH COMPONENT OF CD-DViT ON THREE DATASETS

#	DM	DCAB	CCAB	KD	EHangzhou-RPaviaHR			RPaviaC-DPaviaU			Houston2018-Houston2013		
					OA(%)	AA(%)	$\kappa(\times 100)$	OA(%)	AA(%)	$\kappa(\times 100)$	OA(%)	AA(%)	$\kappa(\times 100)$
1					91.87	96.37	82.94	82.61	82.63	78.23	82.39	83.19	78.21
2	✓				96.30	98.32	91.51	83.74	83.74	79.64	82.67	83.52	79.53
3	✓	✓			96.40	98.36	91.70	84.14	84.15	80.14	82.80	83.67	79.67
4	✓		✓		96.25	98.30	91.40	84.23	84.36	80.25	83.10	83.84	80.02
5	✓		✓	✓	96.48	98.41	91.92	85.97	86.14	82.44	83.33	84.21	80.30
6	✓	✓	✓		96.58	98.47	92.20	84.29	84.33	80.33	82.87	83.79	79.76
7	✓	✓	✓	✓	**96.83**	**98.58**	**92.76**	**86.11**	**86.25**	**82.61**	**83.75**	**84.48**	**80.80**

capabilities of transformer-based methods than CNN-based methods.

2) As depicted in Table VI, CD-DViT exhibits better performance than ViT and SpectralFormer, illustrating the effectiveness of transfer learning.
3) Our proposed algorithm, CD-DViT, still achieves better classification performance than all the compared methods.

Finally, the experiments are conducted on the Houston2018-Houston2013 dataset, and the results of all algorithms are presented in Table VII. The analyses are given as follows.

1) The experimental results in Table VII indicate poor performance of DPyResNet and HybridSN and comparatively better performance of ViT and SpectralFormer, which are better than that of 3-D-SRNet and RPCL-FSL. In addition, CD-DViT exhibits superior performance compared to other transfer learning methods. These results show the strong feature extraction capabilities of the transformer.
2) As depicted in Table VII, the classification accuracy of CD-DViT surpasses that of ViT and SpectralFormer, showing the effectiveness of transfer learning.
3) The results presented in Table VII highlight that the proposed CD-DViT algorithm achieves superior classification performance compared to the other algorithms, demonstrating the effectiveness of combining transformer with transfer learning for HSI classification.

E. Ablation Studies

The proposed algorithm, CD-DViT, primarily contains three key components: the DM, FA branches (where the FA branch further contains DCAB and CCAB), and KD. As shown in Table VIII, seven ablation studies are conducted, and the primary analyses are given as follows.

1) Effects on DM Branch: Observing the results in Table VIII, the DM branch performs better than the baseline on three datasets. Specifically, the OA, AA, and κ of the experiments with the DM branch exceed the baseline by 4.43%, 1.95%, and 8.57%, respectively, on the EHangzhou-RPaviaHR dataset. Besides, on the RPaviaC-DPaviaU and Houston2018-Houston2013 datasets, the increases are 1.13%, 1.11%, and 1.41% and 0.28%, 0.33%, and 1.32%, respectively.

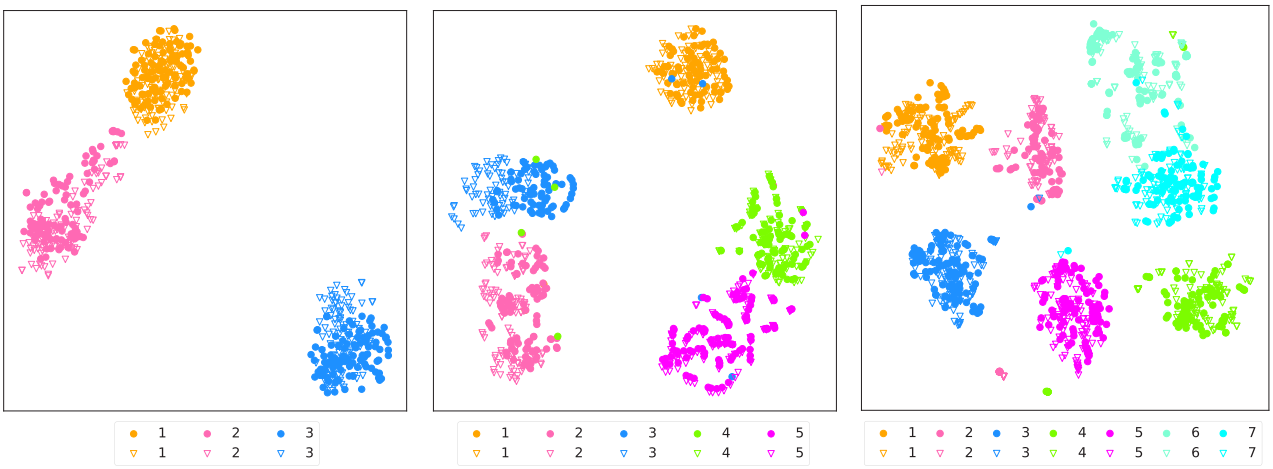
2) Effects on FA Branch: As shown in Table VIII, both DCAB and CCAB with KD can effectively alleviate domain shift and perform well on three datasets. Specifically, the OA, AA, and κ of the experiments with DCAB have increased by 0.10%, 0.04%, and 0.19%, respectively, on the EHangzhou-RPaviaHR dataset. On the RPaviaC-DPaviaU and Houston2018-Houston2013 datasets, the increases are 0.40%, 0.41%, and 0.50% and 0.13%, 0.15%, and 0.14%, respectively. Besides, the experiments on CCAB with KD show improvements of 0.18%, 0.09%, and 0.41% in terms of OA, AA, and κ, respectively, on the EHangzhou-RPaviaHR dataset. The increases are 2.23%, 2.40%, and 2.80% and 0.66%, 0.69%, and 0.77%, respectively, on the RPaviaC-DPaviaU and Houston2018-Houston2013 datasets. Compared to using a single FA block, simultaneously employing DCAB and CCAB with KD could enhance performance among all three datasets. Specifically, the OA, AA, and κ have increased by 0.43%, 0.22%, and 1.06%, respectively, compared to using DCAB solely. On the RPaviaC-DPaviaU and Houston2018-Houston2013 datasets, the increases are 1.97%, 2.10%, and 2.47% and 0.95%, 0.81%, and 1.13%, respectively. Similarly, the increases are 0.35%, 0.17%, and 0.84% in terms of OA, AA, and κ, respectively, on the EHangzhou-RPaviaHR dataset compared to utilizing the CCAB with KD solely. They have increased by 0.14%, 0.11%, and 0.17% and 0.42%, 0.27%, and 0.50%, respectively, on the RPaviaC-DPaviaU and Houston2018-Houston2013 datasets.

3) Effects on KD: It is important to note that in Table VIII, we can observe that solely selecting CCAB without KD and substituting KD loss with cross-entropy loss can hinder the effective training of CCAB parameters, consequently leading to a decline in classification accuracy. Conversely, in experiments where both CCAB and KD are utilized, an enhancement in classification accuracy is evident. Therefore, the combination of CCAB and KD is deemed essential. Specifically, when utilizing CCAB with KD solely, the increases are 0.23%, 0.11%, and 0.52% in terms of OA, AA, and κ, respectively, on the EHangzhou-RPaviaHR dataset. On the RPaviaC-DPaviaU and Houston2018-Houston2013 datasets, the increases are 1.74%, 1.78%, and 2.19% and 0.23%, 0.37%, and 0.28%, respectively. Furthermore, when utilizing both CCAB with KD and DCAB for FAs, the increases are 0.25%, 0.11%, and 0.56% in terms of OA, AA, and κ, respectively, on the EHangzhou-RPaviaHR dataset. On the RPaviaC-DPaviaU and Houston2018-Houston2013 datasets, the increases are 1.82%, 1.92%, and 2.28% and 0.88%, 0.69%, and 1.04%, respectively.

F. Visualization of DM

To show the effectiveness of DM and FA branches, the t-distributed stochastic neighbor embedding (t-SNE) [58] is

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(a) (b) (c)

Fig. 11. t-SNE visualization of features after DM on three datasets. (a) EHangzhou-RPaviaHR. (b) RPaviaC-DPaviaU. (c) Houston2018-Houston2013.

TABLE IX
COMPUTATIONAL TIME (S) AND PARAMETERS (MILLION) OF DIFFERENT DEEP LEARNING-BASED METHODS

method	EHangzhou-RPaviaHR			RPaviaC-DPaviaU			Houston2018-Houston2013		
	Training Time	Inference Time	Parameters	Training Time	Inference Time	Parameters	Training Time	Inference Time	Parameters
DPyResNet	25.22	16.24	22.0616	26.20	0.04	21.9706	26.76	0.72	22.1994
HybridSN	6.02	12.90	5.1205	7.87	0.06	5.1208	8.94	0.55	5.1210
ViT	17.79	10.60	42.6086	18.19	0.04	42.5871	17.11	0.44	42.6440
SpectralFormer	13.11	13.88	0.1565	13.01	0.04	0.1248	14.07	0.59	0.2194
3-D-SRNet	100.78	40.48	0.5105	75.98	0.09	0.5116	77.90	1.36	0.5126
Cycle-AC-GAN	103.43	0.82	1.7772	151.30	0.01	1.3032	217.31	0.04	1.3582
Gia-CFSL	271.45	67.44	0.8622	159.39	0.17	0.8500	258.62	2.33	0.8524
DCFSL	57.65	18.47	4.3512	81.49	0.06	4.3333	94.69	0.87	4.3370
RPCL-FSL	123.26	14.70	4.5300	60.31	0.05	4.4894	76.13	0.64	4.4759
CD-DViT	1226.22	10.88	220.0447	2047.56	0.36	219.9265	2830.56	0.65	219.9978

used to visualize the features after DM in 2-D space. The results of visualization are illustrated in Fig. 11. The circles represent the mapped source features, the triangles represent the target features, and the different colors indicate different land-cover classes.

As shown in Fig. 11, the distributions with identical land-cover classes of source and target features become close after DM. It indicates that the proposed CD-DViT can effectively achieve FA.

G. Comparison on Computational Complexity

We record the computational complexity of different approaches, including DPyResNet, HybridSN, ViT, SpectralFormer, 3-D-SRNet, Cycle-AC-GAN, Gia-CFSL, DCFSL, RPCL-FSL, and our proposed CD-DViT. The training time, the inference time, and the number of parameters for all compared approaches are listed in Table IX. The results show that our proposed approach needs longer training time. This is because the source and target training samples should be paired with identical land-cover classes, significantly increasing the number of training samples. Furthermore, the transformer will map data to a higher-dimensional feature space with multilayer encoders or decoders, increasing the number of parameters. The more significant amount of training samples and parameters will both increase the training time. Besides, it is worth noting that the inference time of our proposed method is similar to other methods, proving the feasibility of CD-DViT in practical application.

V. CONCLUSION AND FUTURE WORK

This article proposed a novel heterogeneous cross-domain HSI classification algorithm named CD-DViT. It contains three branches to address the challenge of small-sample-size problems in ViT-based HSI classification. The DM branch maps source samples to the target domain based on the encoder–decoder architecture. Subsequently, the FA branch contains two blocks based on the cross-attention mechanism to achieve both domainwise and classwise FAs. Furthermore, the TC branch is responsible for classifying the target and mapped source samples. Meanwhile, KD is applied to effectively transfer more transferable information from the target domain to assist in classifying target samples. Experiments conducted on three real-world cross-domain datasets demonstrate that CD-DViT is robust to hyperparameters and outperforms other compared methods.

Despite CD-DViT demonstrating strong performance in cross-domain classification, it is important to note that a constraint of CD-DViT is the need for the source and target domains to share identical label spaces, indicating a requirement for shared land-cover classes. To expand its application

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scope, our focus is on achieving domain adaptation for open-set datasets in our feature work.

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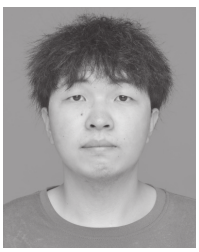
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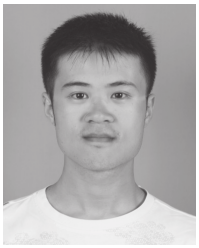
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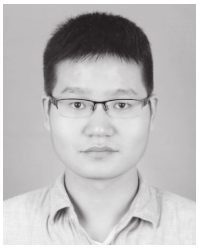
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